Report on the hierarchical Ising opinion model

1 Introduction

Polarization is becoming a common social phenomenon in recent years[1]. After a series of interactions, including exposure under contrary opinions[2] and discussions within group with similar opinions[3], their opinions toward an attitude object tend to polarize further. Such phenomena increase opposition and prevent reaching a cooperation, which is impactful in social issues. Hence, studying the formation of polarization is a crucial topic in sociology.

The journal article ”*The polarization within and across individuals: the hierarchical Ising opinion model*”[4] proposed a HIOM (hierarchical Ising opinion model) to simulate the evolution of polarization in a group of people. The model successfully predicts the polarization effects and gives an explanation base on the cusp catastrophe due to Ising model. The article not only reproduces an empirical case of polarization, but also provides a method to reduce the degree of polarization on an attitude object, which is potentially useful in understanding the polarization phenomenon.

The report aims to look further into the HIOM and its simulation results in order to have a deeper understanding in the implication and mechanism of the model parameters. Several insights were raised and aim to complement the original study, and were inspiring enough to develop new method to reduce polarization by making use of the mechanisms. In addition, the effect of diffusion of opinion by activists is explored.

The report is organized as follows. The HIOM is first explained in detail, then the results produced by the article will be explained and as supplement to the article's findings. At last, the effect of opinion diffusion will be shown and discussed.

2 Methods

The HIOM consists of two levels, the individual level and the interpersonal level. The individual one incorporates some features of the classical Ising model, which is originally about the collective ferromagnetic behavior of a material arising from spin of particles in a lattice[5], to describe the opinion evolution in an agent. While the interpersonal one focuses on the interaction between agents.

2.1 Individual level of HIOM

The individual level of HIOM, referred to as the Ising attitude model in the article, is a network constituted by many interacting attitude element nodes. These nodes could be about feelings, behaviors and beliefs. When an attitude object is raised, a certain combination of nodes that are related to the attitude object will be activated, and the collective behavior of these nodes determines the opinion of an agent.

2.1.1 Glauber dynamics

Glauber dynamics is a kind of Monte Carlo method (simulating complex system by random numbers), which is used to simulate the classical Ising model on a computer. In Glauber dynamics, the energy of Ising model system is given by the Hamiltonian[6, 7]:

$$H(x) = -\sum_{i} \tau_i x_i - \sum_{\langle i,j \rangle} \omega_{ij} x_i x_j \tag{1}$$

Where x_i is the spin of a particle on the lattice network, τ_i is the external field strength and ω_{ij} is the interacting parameter. In some simplification, the interacting parameter and external field strength is often a constant. For Glauber dynamics, the algorithm goes as following[8]:

1. Select a particle randomly.
2. Calculate energy difference $\Delta H = H_{new} - H_{old}$ of the system if the spin of that particle is flipping.
3. Flip the spin according to the probability $P_{flip} = \frac{1}{1+e^{\frac{\Delta H}{kT}}}$, where T is temperature and k is Boltzmann constant.
4. Repeat from step 1.

The Ising attitude model follows Glauber dynamics, and connect it with human psychological activity though four assumptions. Which means the meaning of parameters such as $x_i, \omega_{ij}, \tau_i, \frac{1}{kT}$ have to be explained.

2.1.2 Assumptions

The first assumption is that the nodes can be described by a binary variable, which can be denoted as $x_i \in \{-1, +1\}$ $(i\ in\ 1...n)$, representing a pair of opposite states. The states are related to stances toward an attitude object, for example, in a topic like vegetarian or meat-eater, +1 in a node about feeling (taste of meat) could means tasty, while -1 means the opposite. The sum of x_i defines the opinion of an agent. Hence we have opinion $O = \sum_i^n x_i$, and we interpret positive and negative values as two contrary stance, while the magnitude represent the degree.

The second assumption is that the connections between nodes are symmetric, which means the connection weights ω_{ij} are always the same between a pair of particular nodes no matter the direction. The attitude element nodes can affect each other just like feeling (animals are cute) can affect behavior (eat meat), and two connected nodes always influence each other equally.

The third assumption is that the nodes possess inherent disposition τ_i, which associate with the probability of the node adopting +1 or -1 state without any influence from other nodes. $\tau_i \in (-\infty, +\infty)$ can be projected to a value of probability in $[0, 1]$ by a function in the form $f(\tau_i) = \frac{1}{1+a^{-b\tau_i}}$, where a and b are just some parameters to control the steepness of the function. The reason of choosing whole real range for τ_i is to provide a larger flexibility for HIOM. Dispositions here to Ising attitude model is just what external field strength to classical Ising model[6], which could represent including the social influence (values, ethics) and inborn nature, in situation like a society. To summarize the mean effect of disposition, we have $I = \bar{\tau}$, which is the mean value of τ.

In the fourth assumption, attention A (usually $A > 0$), which is a parameter, was introduced to replace inverse of temperature and Boltzmann constant, $A = \frac{1}{kT}$. Attention tells how sensitive a node will be influenced by other nodes. If attention is high, a node tends to be affected by the network environment easily and show consistency with other nodes. An attention of 0 means completely random behavior of a node, which has no relevance with the network. The attention assumption likely emerged to align with the algorithm of Glauber dynamics, so that the attention is analogous to the inverse of temperature and Boltzmann constant in the spin flip probability of Glauber dynamics. Apart from that, interpreting it as how agents think about an attitude object is also intuitive. If one thinks more on a topic (high A), his nodes flip purposefully and will probably develop a certain stance, if one does not think about a topic (low A or $A = 0$), his nodes flip randomly that he is not likely to have a clear stance.

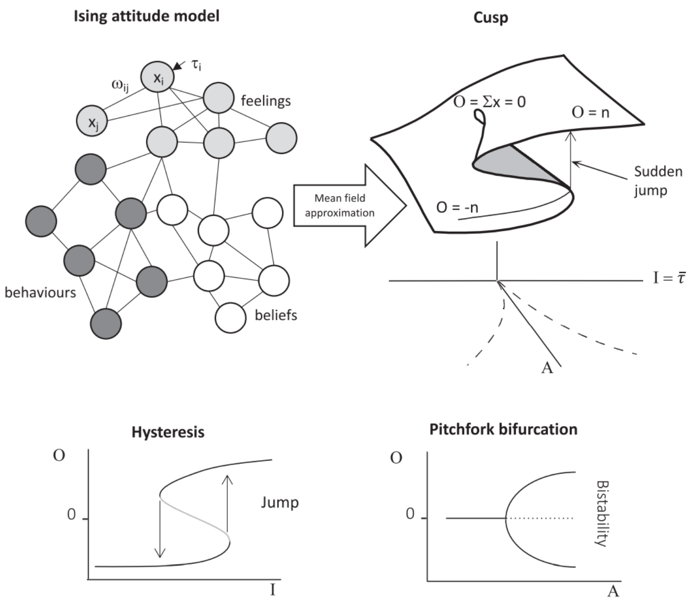

Figure 1: Figure from[4]. The upper part shows after mean field approximation of Ising model, the cusp catastrophe graph can be constructed, the surface is just the equilibrium state of the system. The lower left figure shows hysteresis at a fixed A, while right figure show bifurcation of equilibrium position along fixed I.

2.1.3 Cusp catastrophe

It is easy to understand that the Ising attitude model is simply a psychological reinterpretation of the traditional Ising model by stating some assumptions. Apparently, the Ising attitude model shows similar properties to the classical Ising model. The critical one here is the cusp catastrophe. By mean field approximation, the equilibrium state of Ising attitude model can be described by an equation[9, 4]:

$$O^3 - AO - I = 0 \tag{2}$$

which is a surface in 3D space with O, I, A as parameter (see Figure 1). In fact,Equation 2 is a result telling the stationary point of a potential energy function $V(O) = \frac{1}{4}O^4 - \frac{A}{2}O^2 - IO$ in dynamic system. To find the stationary point, we let $\frac{dV}{dO} = 0$ to obtain Equation 2. To find the rate of change of O, we can consider that the system tends to minimize its potential energy by changing O, the moving direction of O should always point to a local minimum of $V(O)$, so we can have a gradient descent relationship $\frac{dO}{dt} = -\frac{dV}{dO}$ (see Figure 2).

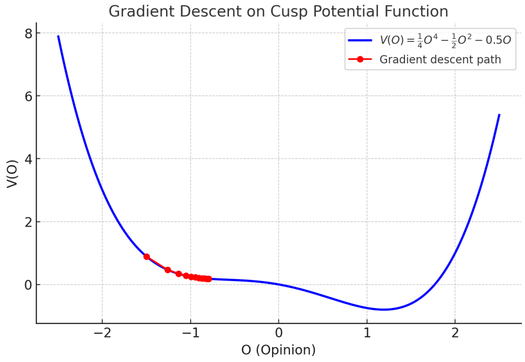

Figure 2: The red dot moving to local minimum at a rate according to slope of $V(O)$ at that point.

So we obtain a differential equation $dO = -(O^3 - AO - I)dt$ describing rate of change of O. For agents at any point in the O, I, A space, they tend to move to the stable equilibrium surface. Note that for area with three stationary position, the middle one is always an unstable stationary position. The movement of agents in the cusp catastrophe is described in Figure 3. We can observe that agents are likely to stick to the equilibrium surface while changing I and A, hence hysteresis and bifurcation of equilibrium appear naturally.

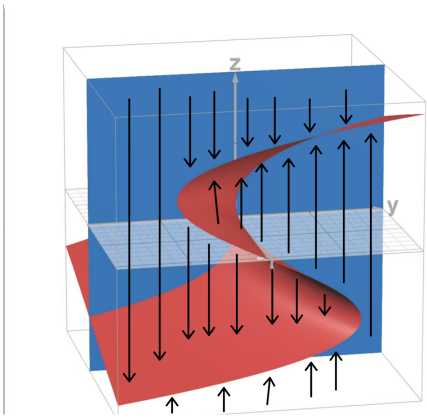

Figure 3: Moving direction of agents in the space.

2.2 The HIOM

The individual level of HIOM concludes the dynamic rules of O, while the interperson level of HIOM gives the update mechanism of I, A. Together with interaction between agents, the HIOM is then complete. In HIOM, agents form a network that can interact with each other, leads to update of their own I and A, which then update their O.

In actual simulation, rate of change of O in each agent $(i...n)$ is modified to have better performance, which is:

$$dO_i = -(O_i^3 - (A_i + A_{min})O_i - I_i)dt + n_o \tag{3}$$

Where A_{min} is a negative number (usually -0.5) introduced to provide linear behavior of O when A is low, and n_o is random noise, equal to the standard deviation of randomly selected numbers in a normal distribution with mean $= 0$.

The HIOM assumes that the probability of an agent begin an interaction depends on his attention, which means agents with high attention are more likely to start an interaction. The probability of choosing an agent i to start an interaction is:

$$P_i = \frac{A_i}{\sum_i A_i} \tag{4}$$

HIOM also assumes that the attention of an agent increase upon successful interaction, and decrease spontaneously with time. The rate of change of attention is:

$$dA_i = -\frac{2d_A}{N} A_i + d_A(2A^* - A_i)\delta \tag{5}$$

Where d_A controls rate of change of A, N is the number of agents in the network, A^* is the equilibrium position of attention of whole network. If the sample size is large, each agent would have lower chance to be selected and increase their attention, hence N is appeared as the denominator in the decay term to make sure the overall attention decay is not too violent in any sample size. In every iteration, if the agent interact successfully, then $\delta = 1$, otherwise, $\delta = 0$.

In each iteration, information I will be updated among agents who involved in an interaction according to:

$$I_{i+1} = rI_i + (1-r)I_j + n_s, \ where\ r = r_{min} + \frac{1 - r_{min}}{1 + e^{-p(A_i - A_j)}} \tag{6}$$

n_s is the noise, which equal to the standard deviation of randomly selected numbers in a normal distribution with mean $= 0$ in each update, and p is a parameter reflecting the persuasive strength during interactions of all agents. $r_{min} \in [0, 1]$ is a parameter imply the inherent resistance to persuasion of all agents. When both agents have the same attention and $r_{min} = 0$, their updated information will be the mean of their previous information, thus we can infer that agents with higher attention tend to drag the information of their opponents towards themselves more.

The individual level and interpersonal level of HIOM can be conclude easily. The former determines update of opinion while the latter determines update of information and attention. Combination of them forms a complete dynamic system that we can observe how agents move in the O, I, A space under different circumstances.

2.3 Measures

Some quantification methods are employed to evaluate the result. The proportion of positive opinion $p(O > 0)$ indicate the distribution of opinions, and the Hartigan D value shows the polarization. Hartigan Dip statistic is used to measure the degree a distribution deflection from a unimodal distribution. To calculate the Hartigan Dip statistic of a distribution, we convert the distribution into empirical distribution function, and compare its largest vertical difference with each possible unimodal cumulative distribution. Then, we choose the minimum largest vertical difference among all unimodal cumulative distribution to be the Hartigan Dip statistic value[10]. So a smaller Hartigan D value means the distribution is more likely to be unimodal.

2.4 simulation

The HIOM algorithm in R code is as follow.

1. Choose a network topology.
2. Initialize parameters $N, A_{min}, A^*, n_o, n_s, d_A, p, r_{min}, t_o, dt, O, I, A$. Commonly, $dt = 0.01$, $A_{min} = -0.5$, and t_o is bounded confidence.

3. Begin iteration.
   (a) Choose an agent according to Equation 4.
   (b) Randomly choose a neighbour as partner for interaction.
   (c) If opinion difference is less than t_o:
      i. Add attention to both agents ($+d_A(2A^* - A_i)\delta$).
      ii. Update information by Equation 6.
   (d) Implement attention decay to all agents ($-\frac{2d_A}{N}A_i$).
   (e) Update opinion on all agents by Equation 3

The code for simulation is available here `https://github.com/ASTRAEL-BAI/HIOM-test`.

3 Extended analysis

There are some interesting observations of the HIOM that give us clues in understanding the model, these include noise, the information parameter and the attention parameter.

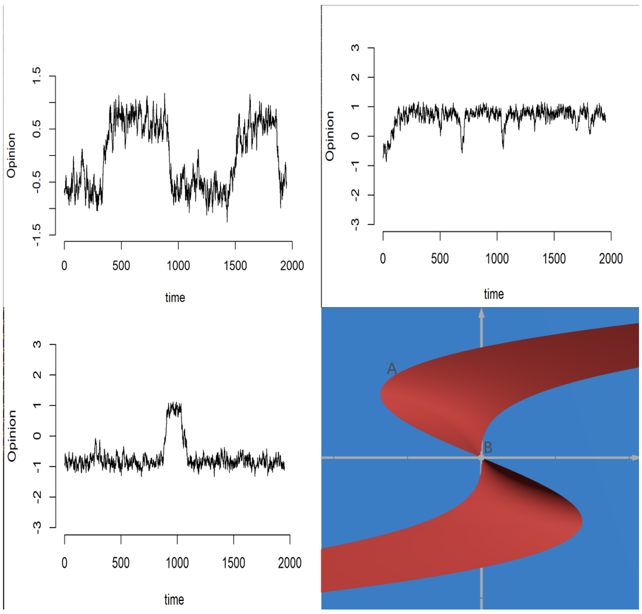

Figure 4: Sudden jump between bistable state due to noise. The top left figure shows sudden jumps of O at $A = 1, I = 0$. The top right figure is $A = 1, I = 0.1$, opinion is much harder to stay in negative O than the positive. The bottom left is $A = 1.3, I = 0$, sudden jumps occur rarer. The bottom right figure illustrate that at position A, which is negative I, the opinion is more likely to get over the unstable equilibrium surface, meaning that it is less likely to stay at positive opinion at that position.

Noise appear in both update of opinion and information. Although opinion usually stay at one of the equilibrium surface in bistable area, sudden jumps to another equilibrium surface is possible if the noise of opinion n_o accumulate to certain extent. Recalling the cusp catastrophe, the unstable equilibrium surface is tilted in the middle, so at fixed A, different I cause the opinion jump differently, meaning that at positive I, opinion stay at positive equilibrium surface more easily. And for fixed I, the jumps occur rarer as increasing A, as a larger accumulated noise of opinion is required to get over the unstable equilibrium surface (Figure 4).

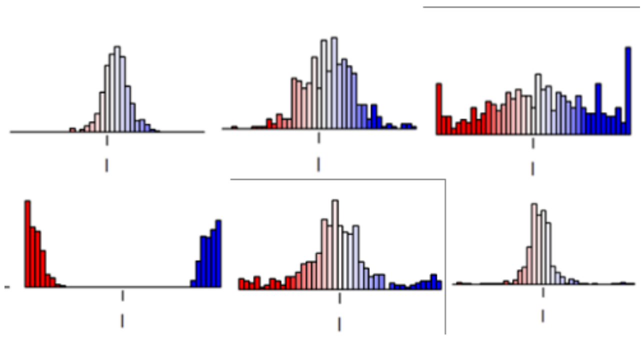

Figure 5: The upper part shows the spread of information distribution of agents solely due to noise (from left to right), no interaction is involved (the accumulation of information at $I = \pm 1$ is because the plot include data points outside the I axis and two ends, it is not polarization, information still spread evenly outside the Iaxis). The lower part shows the convergence of information when agents are initially having information of around ± 0.9 evenly (from left to right), the information distribution converges at around 0 as iterations process. (stochastic block model is used)

For the noise in information update, one interesting phenomenon is that the information of agents will diffuse away along the information axis due to noise during iterations. The spread of information can cause change in stance after enough number of iterations even at $r_{min} = 1$. In addition, we can derive from Equation 6 that after enough iterations that agents keep interacting, the distribution of information eventually converge together, implying an unimodal distribution of information, where the location of unimodal distribution depend on attention distribution. This means agents will have similar information in value, but polarization of opinion is still possible due to bistability (see Figure 5). A trouble of convergence of information is that it confines information distribution in a certain range, meaning that agents will always have similar information when they are living together. However, this cannot explain the occurrence of activists with opposite information. In a society where information eventually converges, no new idea toward the attitude object can be generated. One method to solve the trouble is to introduce some potential activists with high noise in information, so that those potential activists still have chances to reach extreme information.

Attention also shows intriguing behaviors. During simulations, when d_A is relatively small, the overall attention gathered at a value slightly lower than A^*, like there is a peak around A^*. Yet if d_A is relatively large, the peak spread and shift towards 0, and a large fraction of agents having A around 0 (see Figure 6). This result can be explained in two aspects. First, larger d_A leads to fast overall attention decay. Before an agent can interact again, large d_A decrease attention rapidly thus further apart from A^*. Also, when d_A is large, attention of agents can have better chance to go beyond the equilibrium, the larger the d_A, the further attention exceed A^*. This also implies that attention of agents can move around the attention axis in a larger range with higher rate of change of A if d_A is high. Second, based on the selection rule in Equation 4, when d_A is large, after some iterations, a small group of agents will have attention much higher than the non-interacting agents, and the small group of agents will have much higher possibility to be selected and boost their attention even more, resulting in even less chances for non-interacting agents to be selected, the gap in attention widened. This explains the residue at $A = 0$.

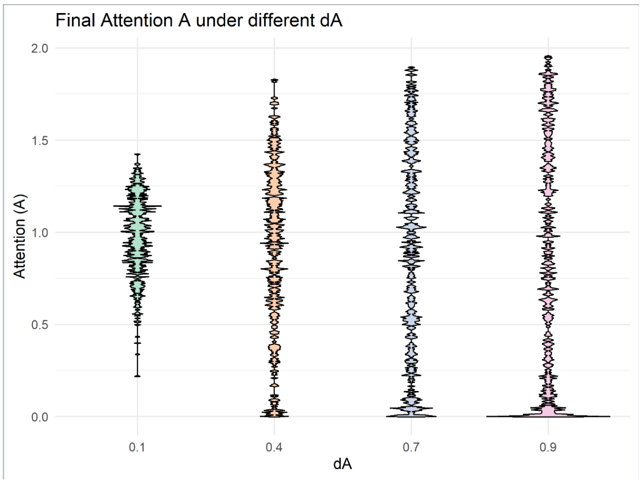

Figure 6: Distribution attention after 15000 iterations for different d_A. When d_A is low, attention gathered around equilibrium. When d_A is high, attention spread away and a fraction of agents having attention of 0.

There are some hypotheses about d_A here. d_A represent the rate of change of attention. Based on Equation 5, it assumes that if one's attention is raised faster, his attention must decay at a higher rate when he is not think about the topic. This resemble some real world situations. Recent studies point out that the attention span of the public toward an object in social media is shortening with steeper rate of change of attention through recent years[11]. As the accelerating information dissemination rate over past few years, we can know and receive information in a short time, that helps us to raise our attention towards a topic more effectively. Hence we can interpret d_A as something related to information dissemination rate in the society, not directly equal (information dissemination also affect I), but higher the information dissemination rate, higher the d_A of the society. So the first hypothesis is a society with high information dissemination rate, agents are more likely to have a shorter attention span on an attitude object, and for that attitude object, most agents eventually lost interest on it quickly and have weak stance on it due to low attention.

Another hypothesis is that when d_A is too large, after some iteration, attention of an agent can become negative by using Equation 5. If d_A is too large, attention can be boosted to a value much larger than A^* after an interaction, thus if $A >> A^*$ continued in second interaction, the term $d_A(2A^* - A)\delta$ become very negative, resulting a negative attention. The trouble is agent with negative attention can not be selected by Equation 4, it will be hard for the agent to escape negative attention unless his partner interact with him. Recall the Glauber dynamics, negative $A = \frac{1}{kT}$ indicates the system tends to increase its energy, leads to opposite spin with neighbour is preferred. In the Ising attitude model, negative A means nodes are more likely to have opposite value with neighbours, resulting in $O \approx 0$. This can be visualized in the cusp catastrophe also. The interpret of negative attention can be quite funny, it is like saying that if you force yourself to think about something too much in a short time, you will get negative attention and ”overload”, feeling so contradictory and giving up to have a stance. Similar phenomenon exists in real world. Exposure under massive information environment often cause cognitive overload and fatigue, leads to avoidance behavior that tends to delay or escape from making a decision or choosing a stance[12, 13]. Although $\frac{1}{kT}$ is not likely to be negative, but the mechanism of A depends on Equation 5, so they are actually not identical, and the interesting one is that negative attention might could has meaning.

4 Results

In this section I will discuss some results produced in the article that are not explained in detail qualitatively.

4.1 Black Pete

According to the article, the Black Pete case show persuasive resistance be cause the uneven change in attention and information for each agent. Two activists with negative opinion, strong (negative) information and attention were intoduced to influence the majority, which is with low attention, positive information and opinion.. We can think of the position of an agent as a position vector $\vec{v}$, and the rate of change of the position vector due to each interaction is $d\vec{v}_i = dI_i\hat{I} + dA_i\hat{A} + dO_i\hat{O}$ according to Equation 6 (dI_i here can mean change of information in one interaction, so it can be $dI_i = I_{i+1} - I_i$), Equation 5 and Equation 3. In each interaction, dA_i only depends on current attention but dI_i can depends on how far an agent at compare to activist. This is because I_{i+1} is a weighted value between two agent's information, so agents near activists can have larger dI_i, while agents far away from activists receive information that is not so extreme, so dI_i is smaller, this decay in dI_i is due to the weighted algorithm in change of information of agents. Recalling the cusp catastrophe in Figure 1, as an agent increasing his attention, the step size and the direction of dI_i decide which equilibrium surface he will finally land.

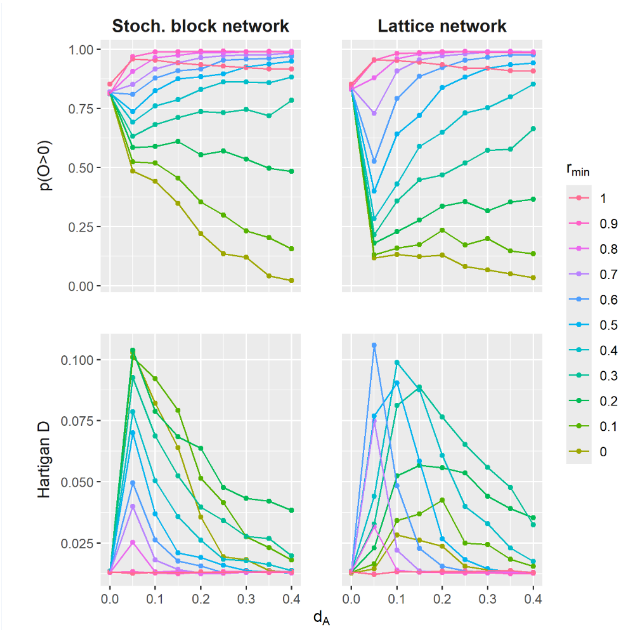

Figure 7: Test of Black Pete case in different parameters. Results here is a bit different with the article one, which is because noise of information was adopted here, while not for the article.

The results of Black Pete under different parameters can be understand qualitatively by using some techniques mentioned in section 3. Different network models are used, but that does not affect the result too much. The network topology can affect the spreading speed of information and attention, but not the fundamental mechanism. In actual quantification of the model, network topology might cause big differences in the work, but these are out of scope here.

In Figure 7, we first look at the upper part. The behavior of $r_{min} = 1$ curve was dominated by noise of information, and no information exchange also avoid converge of information. When

$d_A = 0$, there is no change for attention and information by interaction, but the information noise can spread away and cause some position opinion of agents to switch their stance if they across the cusp line, this is more likely to happen for low attention agents (the majority), so $p(O > 0)$ drops. When there is d_A value, attention can increase so it will be more difficult for agents to across the cusp line due to information noise (they have to move more distance to get across), we can obverse some lines, especially the $r_{min} = 1$, increased suddenly at $d_A = 0.05$. For such small value of d_A, a sharp peak was form at attention equilibrium, but as d_A keep increasing, attention of agents started to move around on the attention axis, and part of the agents having $A \approx 0$. Higher d_A leads to more fraction of agents having relatively low attention, the effect of information noise then affect the result again, causing drop in r_{min} line in $p(O > 0)$. For increasing trend of lines with high r_{min} along d_A, at $d_A = 0$, since attention of the majority cannot be increased, activist drag the information of majority to their side very easily, so we see lower $p(O > 0)$ than line $r_{min} = 1$. As d_A increase, attention of majority increase faster so that they can resist activists' persuasion better, that is the explanation for increasing $p(O > 0)$. So, both d_A and r_{min} can affect the persuasion efficiency by activists. This also explains higher r_{min} can reach $p(O > 0) \approx 1$ faster at same d_A. For low r_{min}, agents can be persuaded by activist easily, this simply explain dropping $p(O > 0)$ along d_A and different r_{min}. For lower part, the Hartigan D is like reflecting $p(O > 0)$ in an other way, the sudden increase at $d_A = 0.05$ is just because activists drag agents to their side easily and low d_A make attention of agents concentrated at A^*, which is overally high that opinion will be more extreme.

4.2 Meat-eating vegetarian

In the this case with bounded confidence t_o, 80% of meat-eaters are low positive information and attention, and 20% vegetarians are having high attention and strong negative information. With introduction of perturbation probability that can reverse the opinion of vegetarian, it is found that polarization can be reduced.

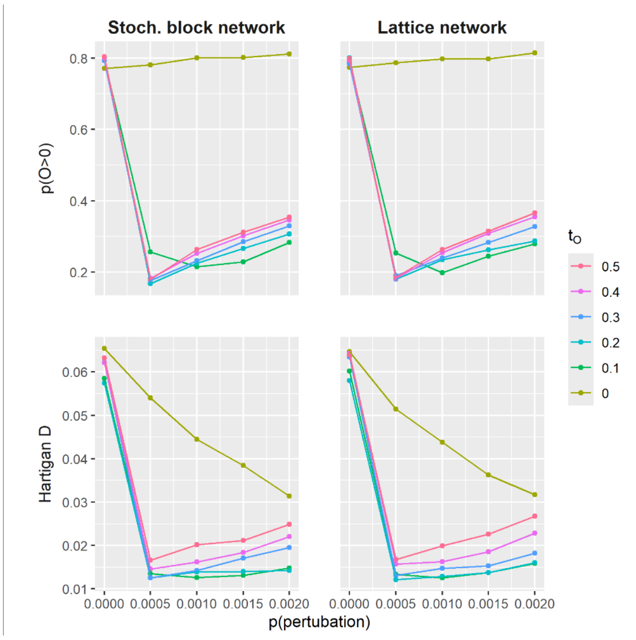

Figure 8: Results of meat-eating vegetarian case under different parameters.

Begin from upper part, when bounded confidence is 0, almost no interact can happen and contribute to change in information and attention. So the increasing trend in $t_o = 0$ line along larger perturbation is just due to more vegetarian reversed their opinion to positive side, nothing more. But we noticed at perturbation $= 0$, $p(O > 0)$ is slightly lower than 0.8, this is just because the noise of information and opinion. Meat-eaters with low attention and information get over the $O = 0$ plane much more easier than vegetarians. For other t_o, agents of same opinions can interact, which avoid their information to spread away due to noise, so at perturbation $= 0$, they are closer to $p(O > 0) = 0.8$. When perturbation introduced, information exchange between opposite opinions is possible that information start to converge (thus depolarized). Since vegetarians have high attention and strong negative information, the converge point will be at area $I < 0$, resulting to that more agents holds negative information and opinions. However, as perturbation probability increases, larger fraction of agents at negative opinion reverse their opinion to positive, leads to an increase in $p(O > 0)$, this also explains the increasing Hartigan D. For lower part, the Hartigan D just behave according to $p(O > 0)$.

4.3 Depolarization by d_A

As discussed in section 3, larger d_A allows agents move around in the attention axis faster and in larger range, and leaving part of agents having low attention. Depolarization is possible if we first converge information of all agents, then by making use of high d_A, let agents go to the low attention area and come back again, or just leave them at linear opinion area (which is at $A < 0.5$ for $A_{min} = -0.5$), they will have opinions based to their information, even in bistable area.

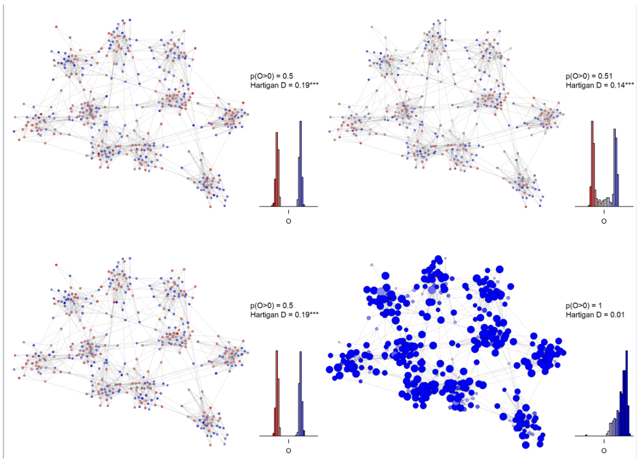

Figure 9: Depolarization by large d_A. Situation on the top has $d_A = 0.1$, while the bottom one has $d_A = 0.9$, other parameters are completely same. The size of nodes just represent the value of attention.

With parameter $n_s = 0.0001, p = 2, r_{min} = 0.1, n_o = 0.01, A^* = 1, A_{min} = -0.5$, for top scenario, $d_A = 0.1$, for bottom one, $d_A = 0.9$, we can observe that polarization persist when d_A is low, but if d_A is high, polarization disappear and an unimodal distribution of opinions produced. This result shows that in a society with high d_A (eg. fast information dissemination), people are more likely to have similar opinions with others, following what other says, and have relatively low attention to attitude objects (do not think much on a topic). This might be a possible solution to reduce polarization in real world, not only because studies show that attention of people decrease if there is too much information[13], but also we can interpret d_A in real world as something related to information dissemination that we can manipulate.

4.4 Diffusion pattern of activist

In the Black Pete case, some parameters can affect the range that an activist can successfully persuade an agents. We can look more into it.

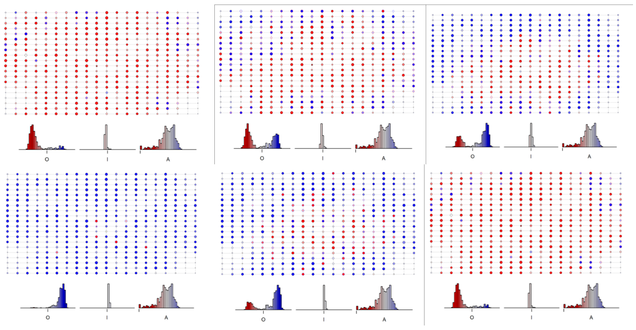

Figure 10: Parameters of $n_s = 0, d_A = 0.1, n_o = 0.01, A^* = 1, A_{min} = -0.5$ are used. The upper part shows when $p = 2$, $r_{min} = 0, 0.2, 0.4$ (left to right). The bottom shows at $r_{min} = 0$, $p = 1, 1.5, 2$ (left to right).

From the above results, we can see a qualitative trend as p and r_{min} increase, however, a quantitative method to indicate the result is not easy. We cannot just simply think about the distance as not only the shape of diffusion is irregular, but also the emergence of agents with same stance with activist happen a bit randomly (red spot appear in somewhere surrounded by blue spot), making the diffusion is not so continuous. But at least we are able to observe a trend and see parameters that affect the diffusion range visually. As r_{min} and p increase, activists are more difficult to affect other agents due to that they cannot influence their information effectively.

5 Discussion

As discussed in section 3, the inevitable convergence of information might be a problem. Other than having larger information noise, some modification on agents or interacting mechanisms can be further investigated. For exmaple, interaction between agents with same stance can move their information to a more extreme stance. d_A was discussed also, we mentioned that it is possible to have some relationship with information dissemination in real world. However, we still do not have a method to quantify a d_A in real world (we do not know how fast information dissemination is in real world if $d_A = 0.6$.), so empirical researches and quantification methods is needed to make the HIOM more reliable, so that we know how a society is like specifically if $d_A = 0.4$. Apart from d_A, we also need an explicit meaning in real world about other parameters, like A^*.

Although HIOM already show some interesting features like the cusp catastrophe, more modification can be made into the model based on emipical researches. Such as the network topology, connection between agents may change frequently, or agents can interact with each other simultaneously. The model restricted agents interact with each other one after another, interaction between multiple agents is impossible. But in real situation, when a group of interact together, group polarization can occur that push their opinion into a more extreme degree[14]. Another aspect is about individual agent. There is lack of evidence show that the process of forming an opinion equals solving for least energy as Ising model. The thing is that we do not know how human mind works yet, and the proof is a really hard work, so we can now just pretend it as an assumption.

For the diffusion of activist, we can observe a range affected by activist visually, but quantifying it is need more research and methods, which we expect the work can be done in the future.

6 Conclusion

In this report, we try to understand the HIOM and analysis some parameters and mechanisms of it, including effect of noise, dynamics of I and A. By looking further into these mechanisms, we raised a method to reduce polarization and tried to view the diffusion effect due to activists in Black

Pete case. We also gave qualitative explanations to some results from the original article in order to have deeper understanding in the HIOM.

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